Cooling Equipment Sensor-Fault Detection

Deep Learning project summary | Section B
Manipal University Jaipur | 3 November 2026

Objective and project development

We first built a detector for outdoor-air temperature sensor bias, targeting one sensor location. We then expanded the project to four sensor fault targets: cold-deck temperature, hot-deck temperature, cold-deck pressure and hot-deck pressure. The LSTM distinguishes these four targets from healthy operation; each scenario contains one faulty sensor. The extension uses a separate simulated AHU dataset.

Data and method

The LBNL simulated dual-duct AHU dataset supplies 17 parallel annual scenarios. The model uses 26 inputs (21 measurements and five control commands) and histories of 30 consecutive operating one-minute readings. Matching calendar weeks remain together across scenarios. Cleaning, scaling and feature selection use training data only. A 32-unit LSTM and 16-unit classification head (8,293 parameters) are trained with seed 42. Validation freezes the model and alarm thresholds before the reserved test; a five-input alternative is also evaluated.

Reserved test results

Five-class accuracy	92.87%	Macro F1	89.36%
Any-fault recall	93.05%	Healthy false-positive rate	2.10%

The test contains 463,058 scored minutes. Correct-sensor recall ranges from 87.52% to 95.88%. **The 1% healthy false-positive goal remains unmet:** 580 of 27,640 healthy readings raise an alarm. The five-input alternative performs worse. No thresholds are retuned on the test.

Deliverables and limits

The code ZIP includes Python source, a Streamlit dashboard, saved weights and preprocessing, replay examples, tests and an executed notebook. The saved demo runs offline after dependencies are installed; the 1.7 GB raw dual-duct archive downloads separately for retraining. Evidence covers one simulated unit, known single faults and one seed. Simultaneous faults, physical-equipment reliability and future-failure forecasting are outside this study.

Proposed team roles and responsibilities

Team member	Responsibility and deliverables
Rishabh Srivastava 2430010149	**LSTM input preparation and feature experiments.** Prepare 30-step LSTM sequences and training-only scaling; validate operating-window eligibility and calendar separation. Train and compare full-input and five-input LSTM variants, and document the feature-selection results. *Deliverables:* prepare_multisensor.py; input-selection study; LSTM comparisons
Mitul Seth 2430010099	**LSTM architecture and four-sensor expansion.** Led the expansion from single-sensor detection to four-sensor diagnosis, designed and trained the LSTM, configured class outputs and alarm thresholds, and integrated dashboard inference. Coordinated end-to-end testing and reproducibility checks. *Deliverables:* multisensor.py; validation policies; dashboard/app.py; verify_multisensor.py
Madhav Mittal 2430010096	**Single-sensor LSTM, training experiments and evaluation.** Develop the single-sensor LSTM; compare epoch budgets and healthy-class weighting, select validation checkpoints, and analyse learning curves and sensor errors. *Deliverables:* train_hvac.py; hvac_inference.py; train/refine scripts; validation histories; evaluation notebook

Sources: LBNL dual-duct AHU dataset, DOI 10.25984/1881324; saved project evidence in results/multisensor_v2/test_summary.json and evidence_summary.json.